ADVANCED MACHINE LEARNING

Assignment 2: Convolution
Convolutional Neural Networks for Cat and Dog Classification

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1. Introduction

Image classification is one of the practical applications of deep learning, where a model learns to recognize patterns in images and assign them to different categories. In this assignment, I explored how convolutional neural networks (CNNs) classify images of cats and dogs, with particular attention to the amount of training data and the type of network used.

I conducted six experiments using two approaches: a CNN developed from scratch and a pretrained MobileNetV2 model. Each approach was tested with 1,000, 3,000, and 6,000 training images, while the validation and test datasets remained unchanged. My objective was to determine how increasing the training data affects classification accuracy and whether transfer learning provides an advantage over training a network from the beginning.

The report presents the main experimental results, compares model performance, and discusses the practical lessons learned from the experiments.

2. Dataset and Methodology

2.1 Dataset Preparation

For this assignment, I used the *Dog and Cat Classification Dataset* available on Kaggle. After checking the image files, I identified 24,998 readable images, equally divided between the cat and dog categories.

To examine the effect of training dataset size, I prepared three balanced training subsets containing 1,000, 3,000, and 6,000 images. The smaller subsets were included within the larger ones, allowing me to increase the training data while keeping the image selection consistent.

I reserved 500 images for validation and another 500 for testing, with equal numbers of cats and dogs in each group. These images were kept separate from the training datasets and remained unchanged across all six experiments. All images were resized to 150 × 150 pixels and converted to RGB format before being processed by the models.

2.2 Model Development

I compared two approaches to image classification.

CNN Trained from Scratch: I developed a CNN using three convolutional layers with 32, 64, and 128 filters, followed by max-pooling, global average pooling, and a dense classification layer. I used image augmentation, dropout, and early stopping to reduce the risk of overfitting. This model learned its visual features directly from the training images.

Pretrained MobileNetV2: For the second approach, I used MobileNetV2 with weights pretrained on ImageNet. I kept the pretrained feature-extraction layers frozen and trained a new classification layer for cat and dog prediction. This allowed the model to reuse existing visual features rather than learning everything from the beginning.

Both approaches were evaluated at the three training dataset sizes. I used accuracy and binary cross-entropy loss to compare their performance on the same 500-image test dataset.

Experiment	Training Images	Validation Images	Test Images
Small Dataset	1,000	500	500
Medium Dataset	3,000	500	500
Large Dataset	6,000	500	500

Table 1. Dataset sizes used for both CNN architectures. Validation and test datasets were held constant across all experiments.

Both models were trained using the Adam optimizer and binary cross-entropy loss. The scratch CNN used a learning rate of 0.001 with a maximum of 30 epochs, while MobileNetV2 used a learning rate of 0.0001 with a maximum of 20 epochs. Early stopping was included to monitor validation loss and restore the best model weights when applicable.

3. Experimental Results

I evaluated six CNN models to examine how training dataset size and network architecture affect image classification performance. Three models were trained from scratch, while the other three used pretrained MobileNetV2. All models were evaluated on the same 500 test images to maintain a consistent comparison.

Table 2 summarizes the test accuracy and test loss for each experiment.

Table 2. Performance Comparison of CNN Models

Model Architecture	Training Images	Test Accuracy (%)	Test Loss
Scratch CNN	1,000	61.80	0.6696
Scratch CNN	3,000	72.00	0.5575
Scratch CNN	6,000	75.00	0.5169
MobileNetV2	1,000	95.40	0.1244
MobileNetV2	3,000	96.00	0.0982
MobileNetV2	**6,000**	**96.80**	**0.0827**

3.1 Key Findings

The results show that increasing the training dataset improved the performance of both CNN architectures. For the CNN trained from scratch, test accuracy increased from 61.80% to 75.00% as the number of training images increased from 1,000 to 6,000. This represents an improvement of 13.20 percentage points.

The pretrained MobileNetV2 models performed substantially better at every training size. Even with only 1,000 images, MobileNetV2 achieved 95.40% test accuracy, outperforming the scratch CNN trained using 6,000 images.

Increasing the pretrained model's training dataset from 1,000 to 6,000 images improved test accuracy from 95.40% to 96.80%. Although the improvement was smaller than that observed for the scratch CNN, the pretrained models consistently maintained high accuracy and low test loss.

The best-performing model was pretrained MobileNetV2 trained with 6,000 images, achieving 96.80% test accuracy and a test loss of 0.0827.

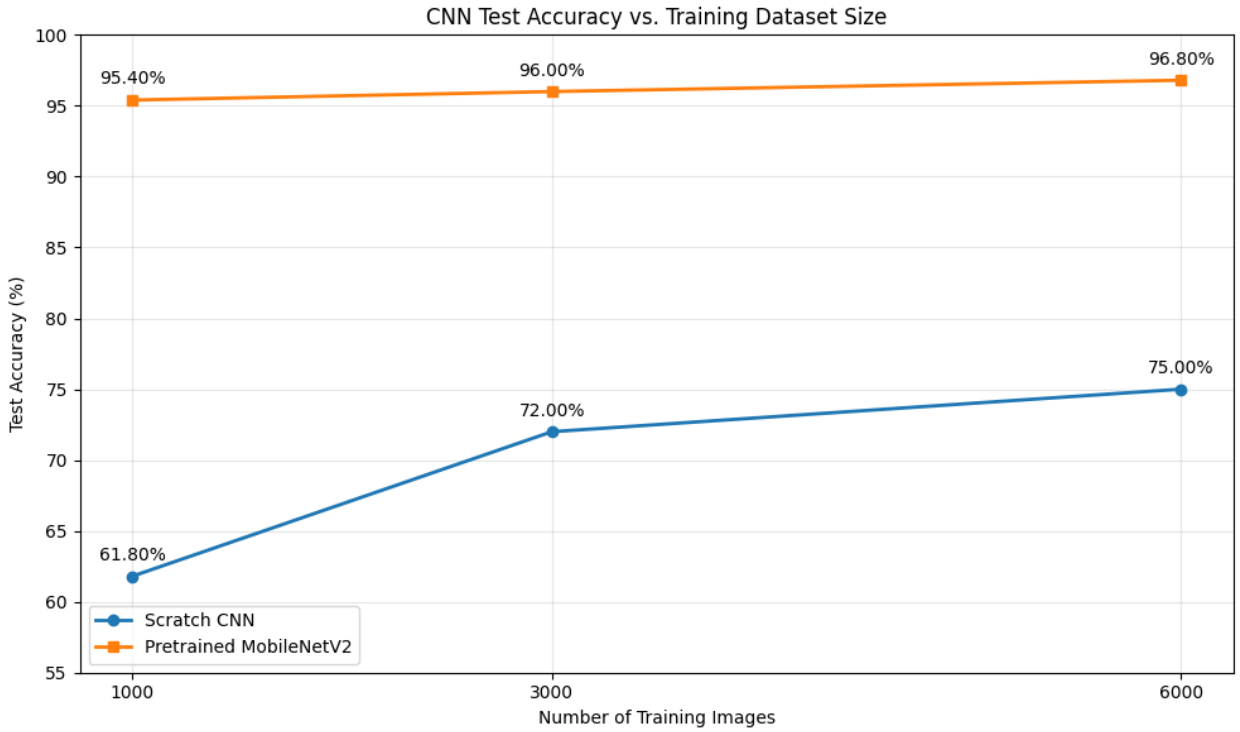


Figure 1. Test accuracy comparison between the scratch CNN and pretrained MobileNetV2 across three training dataset sizes.

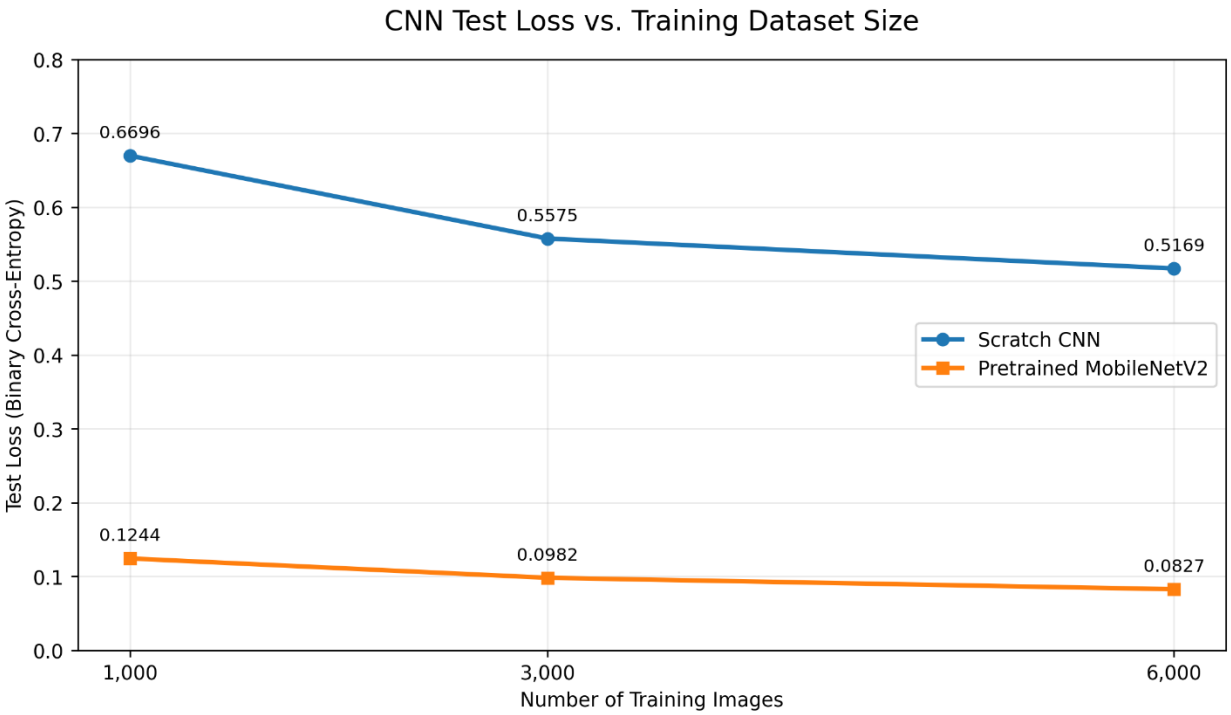


Figure 2. Test loss comparison between the scratch CNN and pretrained MobileNetV2 across three training dataset sizes.

Figure 2 shows that test loss decreased as the training dataset increased for both CNN architectures. MobileNetV2 consistently achieved lower test loss than the scratch CNN across all three experiments. These findings support the accuracy results and further demonstrate the advantage of transfer learning in this classification task.

4. Discussion and Interpretation

4.1 Impact of Training Dataset Size

One of the main findings from my experiments was that increasing the training dataset improved the performance of both CNN architectures, but the improvement was not equally large.

The CNN trained from scratch showed the greatest improvement, gaining 13.20 percentage points in test accuracy when the training dataset increased from 1,000 to 6,000 images. This suggests that the scratch CNN depended heavily on the available training examples because it needed to learn image features directly from the dataset.

In comparison, pretrained MobileNetV2 improved by only 1.40 percentage points over the same range. I found this particularly interesting because it suggests that a pretrained network can achieve strong

performance without requiring as much task-specific training data. The smaller improvement also indicates that simply increasing the dataset may provide diminishing benefits once a model is already performing well.

4.2 Why Transfer Learning Performed Better

The most noticeable difference between the two approaches was the advantage of transfer learning.

The scratch CNN had to learn how to identify useful visual patterns, such as shapes, textures, and distinguishing features, using only the cat and dog training images. MobileNetV2, however, already had feature-extraction layers trained on ImageNet, which provided a stronger starting point.

This difference was especially clear in the smallest training experiment. With 1,000 images, MobileNetV2 outperformed the scratch CNN by 33.60 percentage points. Even the pretrained model using 1,000 images performed better than the scratch CNN using 6,000 images.

Based on these results, I would prefer transfer learning for a similar image classification problem, particularly when the available training dataset is limited. It can provide strong predictive performance without requiring every visual feature to be learned from the beginning.

4.3 Overfitting and Model Generalization

Overfitting was an important consideration because a model can perform well on training images but struggle with images it has not previously seen.

To reduce this risk, I used image augmentation, dropout, and early stopping in the experiments. The pretrained models also used frozen feature-extraction layers, which limited the number of parameters being updated during training.

The pretrained models achieved high validation and test accuracy, and their training and validation results remained relatively close during the later epochs. This suggests that severe overfitting was not evident in these experiments. However, good performance on one test dataset does not guarantee the same results on every collection of images. Testing the models on additional datasets would help determine how consistently they generalize.

4.4 Practical Implications

These experiments showed me that model selection can be just as important as increasing the amount of training data.

If I were developing a similar classification system with limited data, I would begin with a pretrained CNN rather than immediately building a network from scratch. I would then evaluate whether adding more training examples provides enough improvement to justify the additional data preparation and training effort.

For this assignment, pretrained MobileNetV2 with 6,000 images produced the strongest results. However, the model using only 1,000 images also achieved high accuracy, making it a useful option when collecting or labeling more data is difficult.

5. Conclusion and Recommendations

This assignment helped me understand how training dataset size and CNN architecture influence image classification performance. Across the six experiments, increasing the number of training images improved accuracy for both approaches, although the effect was much greater for the CNN trained from scratch.

The most important finding was the advantage of transfer learning. Pretrained MobileNetV2 consistently outperformed the scratch CNN, even when trained with fewer images. The best-performing model was MobileNetV2 using 6,000 training images, achieving **96.80% test accuracy and a test loss of 0.0827**.

Based on these findings, I would recommend starting with a pretrained CNN for similar image classification tasks, especially when training data is limited. Additional training images can still improve performance, but their benefits should be evaluated against the time and resources required to prepare them.

5.1 Limitations and Future Improvements

Although the results were encouraging, the experiments had some limitations. Each model configuration was trained only once, and performance was evaluated on a fixed test dataset of 500 images. Results may differ with other random seeds, image collections, or more challenging classification tasks.

Although 6,000 training images produced the highest test accuracy among the three sample sizes examined, this does not necessarily mean it is the optimal dataset size. Testing additional sample sizes could help identify where further increases in training data provide only small improvements.

In future work, I would repeat the experiments with different random seeds, evaluate the models on additional datasets, and explore fine-tuning selected MobileNetV2 layers. These changes could help determine whether the observed improvements are consistent and whether classification performance can be increased further.

6. References

Jikadara, B. (n.d.). *Dog and cat classification dataset*. Kaggle. https://www.kaggle.com/datasets/bhavikjikadara/dog-and-cat-classification-dataset

Sandler, M., Howard, A., Zhu, M., Zhmoginov, A., & Chen, L.-C. (2018). *MobileNetV2: Inverted residuals and linear bottlenecks*. Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. https://arxiv.org/abs/1801.04381

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